ReadMe
HVAB PIV Data
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This document summarizes the test conditions for which Particle Image Velocimetry (PIV) images were acquired during the HVAB hover test. It also provides brief descriptions of the image processing procedures, the final vector field data set, and the spreadsheets containing the trajectory and evolutionary characteristics of the tip vortices (core size, peak swirl velocity, and total vortex circulation). A few plots for reference are also provided.

The text in this Readme file is largely extracted from the following references for 1) details on PIV experimental set up, data acquisition and processing procedures, and 2) details on the HVAB rotor setup, trim conditions, and facility installation:

1. Ramasamy, M., Heineck, J.T., Yamauchi, G., Schairer, E., and Norman, T., "Comprehensive Aerodynamic Analysis of PIV Measurements in the NFAC 80- by 120-ft Test Section Towards Understanding HVAB Hovering Rotor Characteristics," Vertical Flight Society's 80th Annual National Forum and Technology Display, Montreal, Canada, May 7—9, 2024.

2. Norman et al., "Fundamental Test of a Hovering Rotor: Comprehensive Measurements for CFD Validation," Vertical Flight Society's 79th Annual Forum & Technology Display, West Palm Beach, FL, USA, May 16-18, 2023.

Summary of PIV Test Conditions

Table 1 provides a summary of the run numbers, test conditions, and wake ages for which PIV were acquired during the HVAB test.

Run	M_{tip}	Collective pitch, q (deg)	Blade number	Wake age, ζ deg
92	0.65	8	B1	3, 7, 10, 15, 30, 60, 93, 105, 120, 150, 183,195, 210, 240, 273, 285, 300, 330
95	0.65	10	B1	3, 15, 30, 60
95	0.65	12	B1	3, 5, 10, 15, 30, 60, 93, 105, 120, 150, 195, 210, 240, 273, 285, 300, 330
95	0.65	14	B1	3, 15, 30, 60, 93

Table 1. Test Matrix for PIV image acquisition

While flow field data is available for all the cases listed in Table 1, tip vortex properties (location, core size etc.) are provided only for those in black fonts. The cases with red font did not yield reliable values for various reasons: (1) blades blocking the line of sight for either of the cameras, (2) tip

vortices not discernible at older wake ages to provide statistical evaluation, and/or (3) blade interacting with tip vortex. Tip vortex properties are not available for identical wake ages among various collective pitch conditions because of the different tip vortex convection velocities and the associated variation in the blockage of line of sight caused by the blades. Total vortex circulation is provided only for blade 1 tip vortices.

Zero-degree wake age is assumed when the PIV laser plane connects the rotor shaft with the trailing edge of Blade 1 (while the rotor is spinning at full RPM and, thus, accounting for blade lag). Note that since the HVAB is a four-bladed rotor system, the data in Table 1 can be resolved to extract tip vortex properties for blades 2 to 4 also.

PIV Test Setup and Processing Procedures

Phase-locked, stereoscopic, 3-component PIV measurements were made by synchronizing two 16-MP cameras and two pulsed lasers (1.4 J/pulse) with the rotor rotational frequency. Blade-to-blade differences were studied by comparing the flow field acquired at the same wake age of different blades. Images were acquired at 1Hz (i.e., an image pair/camera about every 20 rotations of the rotor). Two hundred realizations were acquired for each test case.

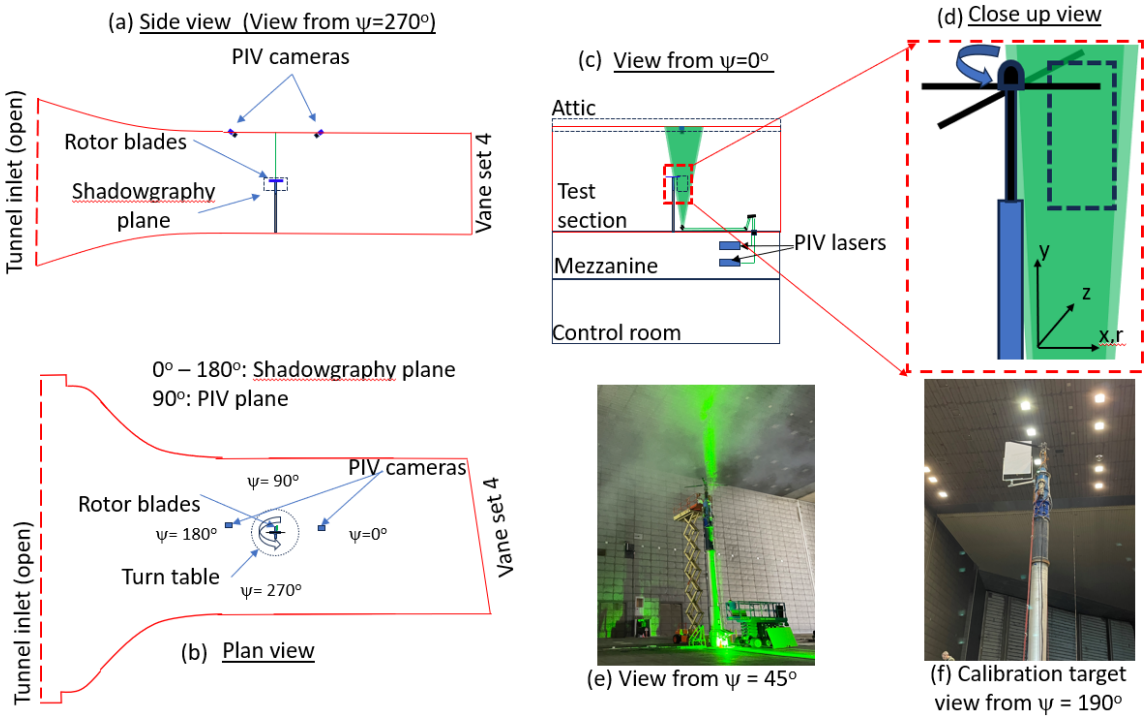


Figure 1. Experimental setup with coordinate system

An approximate scaled-down schematic of the experimental set up in the NFAC test section is shown in Fig.1. Three sides of the test section are closed (no airway); the tunnel inlet was open for the entire HVAB test campaign, including during PIV image acquisition. The rotor coordinate system and azimuthal positions are also shown. The PIV measurement plane was along the 90 deg azimuth.

For comparison, shadowgraphy image plane covered both sides of the rotor along the 0-180 deg azimuth (see Fig. 1a and 1b).

The laser heads were placed in the Mezzanine (one floor beneath the test section as shown in Fig.1c); the beams were driven independently and combined at the region of interest (ROI). After passing through 3x beam expanders to reduce divergence, the beams were vertically propagated into the test section through a 9-inch hole in the mezzanine ceiling into the test section. The 25-mm beams entered the test section and were immediately turned 90 degrees using mirrors placed about 1-ft above the floor. The beams traveled 30-ft parallel to the floor and were again turned 90 degrees by another pair of mirrors (Fig. 1e). The beams then reached the base of the rotor stand where two mirrors directed the beams upwards toward the rotor. A series of plano-concave-convex lenses were placed after the last mirror to generate light sheets of desired dimension. The entire mirror/lense assembly was mounted on an optical table and included several fine adjustment controllers (translation stages/goniometers) that enabled the best overlap between the sheets.

The cameras were mounted in the test section attic on either side of the light sheet in the forward-scattering mode (Fig. 1a). The cameras were fitted with 400-mm focal length lenses and a 532-nm notch filter to reduce ambient light. Dual-axis Scheimpflug mounts were installed for compound off-axis focusing. The cameras used full-frame sensors (4872 x 3248 pixels); the size of each square pixel was 7.4 μm. A custom made, single-plane, double-sided target was mounted on an adjustable stand and was placed parallel to the laser light sheet during system calibration. The width of the target plate spanned from the root cut-out to 1 inch beyond the tip of the rotor blade (see Fig. 1f). The target was positioned such that the laser plane bisected the middle (along the thickness) of the calibration target. Self-calibration (using the first of the two images from both the cameras) was performed to account for the focal plane difference (laser plane vs. faces of the calibration board). The average uncertainty in converting the raw world coordinates to image plane coordinates after self-calibration was about one-tenth of one pixel.

The time between two laser pulses (Δt) was varied based on the collective pitch as well as the maximum displacement of the smoke particles found near the tip vortices. The goal was to achieve 4-5 pixel displacement; this resulted in a Δt of about 50-μs for the 8-deg collective pitch and 45, 40, and 35 μs for 10, 12, and 14 deg collective pitches, respectively. After every acquisition, correlation evaluation was conducted using a 32-pixel window to assure good signal-to-noise ratio in the resulting vector field.

Several pre-test evaluations were conducted by placing seeders at different locations on the floor and in the attic towards achieving uniform seeding density. After several attempts, especially those conducted when the rotor was running at the desired thrust condition (for other measurements such as transition or shadowgraph), ideal positions for the placement of the seeders were identified at 90, 180, and 300-deg azimuths around the rotor and about 50-ft from the base of the rotor stand. The nominal size of the smoke particles produced by the MDG seeders used in the study was about 0.9 micrometers.

The images were processed using DaVis 8.4 commercial software. A multi-step process that began with the initial interrogation window size of 64x64 pixels was reduced to a final window size of 32x32 pixels with 75% overlap. Each interrogation window produces N^2 correlation values, where N is the size of each window in terms of pixels. A vector is considered valid when the ratio of the highest two

peak correlation values is larger than a threshold set by the user. In the present study, the ratio was set to 1.25 and post-processing removed vectors that fell below the threshold. Smoothing or replacing the removed vectors through interpolation was not implemented. The removed vectors created vector voids in the individual realizations.

Post Processing for Lack of Seed (Smoke Particles)

The efforts to uniformly fill the ROI with the seed particles, especially when the average time to set up and acquire data required about 5 minutes, was not practical. Consequently, many realizations were expected to be partially filled, if filled at all, with smoke particles or overfilled. Similarly, the centrifugal and Coriolis forces acting near the center of the tip vortex produces seed void as well. As PIV relies on statistics-based pattern tracking, neither high-density smoke that saturates the images nor low-density smoke that fails to produce a pattern is useful. Controlling the seed particle density and directing the smoke to the desired location were two of the biggest challenges in the current experiment.

Two post-processing algorithms were developed to remove the vector field outliers caused by lack of smoke in the desired ROI. These are: (1) Modal outlier method or MOM, and (2) Projection on phase-average or POP. Detailed description on the development and application of the models are provided in Ref. 1 (listed at the top of this ReadMe report). Seed void at the center of the vortex (and the resulting vector void) is an issue when extracting tip vortex properties, such as spatial location relative to the blade tip, vortex size, peak swirl velocity etc. However, such issues were overcome by applying a least squares curve fit to the data set, which is also described in Ref. 1.

Reduced PIV Data and Description of Data Files

The data reduction procedures were applied to all PIV data acquired as part of the HVAB program. Full vector fields were produced for each condition outlined in Table 1 and tip vortex properties (core size, swirl velocity, total circulation, and trajectories) were extracted for most conditions (see Ref. 1 for a full description of this procedure). Descriptions of the resultant data file formats are provided in the following sections.

Vector field dataset

All vector field data are provided together in the file “VectorFielddata.zip”. This zipped archive includes mean and standard deviation files for all test conditions and wake ages identified in Table 1. For files included in the archive, the naming convention is similar to that used for other HVAB data types, **RxxMxxxThxxBxWakeagexxx**, where

Rxx-Run number (R92 – corresponds to 8 deg case, R95 corresponds to 10, 12, and 14 deg cases)

Mxxx-Mach number (M650 corresponds to tip Mach number of 0.65)

Thxx - Collective pitch angle (Th08 corresponds to 8 deg pitch)

Bx – Blade number (B1 is blade 1)

Wakeagexxx- angle between blade trailing edge and the PIV measurement plane (Wakeage030 corresponds to 30 deg wake age).

The following list describes the symbols used in each of the vector field data files:

r/R – radial distance from the shaft axis

y/R – axial distance from the tip path plane

vrad – radial velocity in m/s

vaxial – axial (downwash) velocity in m/s

w – chordwise velocity (deficit or excess) in m/s

where R is the radius of the rotor blade. Even though 200 realizations were obtained for each wake age, the application of the MOM and POP filters, described in Ref. 1 to remove the "no-smoke" outliers, reduced the number of realizations used in calculating the statistical means and standard deviations (and is different among the test cases). The provided means and standard deviations represent the experimental observations after removing the no-smoke outlier data sets.

Tip Vortex Properties

The extracted tip vortex properties (see Ref. 1 for the extraction procedure) are provided in two files: (1) CoreSize_SwirlVelocity.xlsx, and (2) TipVortexTrajectory.xlsx. The following list describes the symbols used in the core property file:

r_c – core radius of the tip vortex in mm

$V_{\theta max}$ – peak swirl velocity in m/s

σr_c – standard deviation of the measured core radius

$\sigma V_{\theta max}$ – standard deviation of the measured peak swirl velocity

Γ – Total vortex circulation, m^2/s

B1, B2, B3, B4 – Blades 1 to 4, respectively

This list describes the symbols used in the vortex trajectory file:

ζ – wake age in degrees

r_v/R – non-dimensionalized radial location of the tip vortex

y_v/R – non-dimensionalized axial location of the tip vortex

std r_v/R – standard deviation of the radial location of the tip vortex

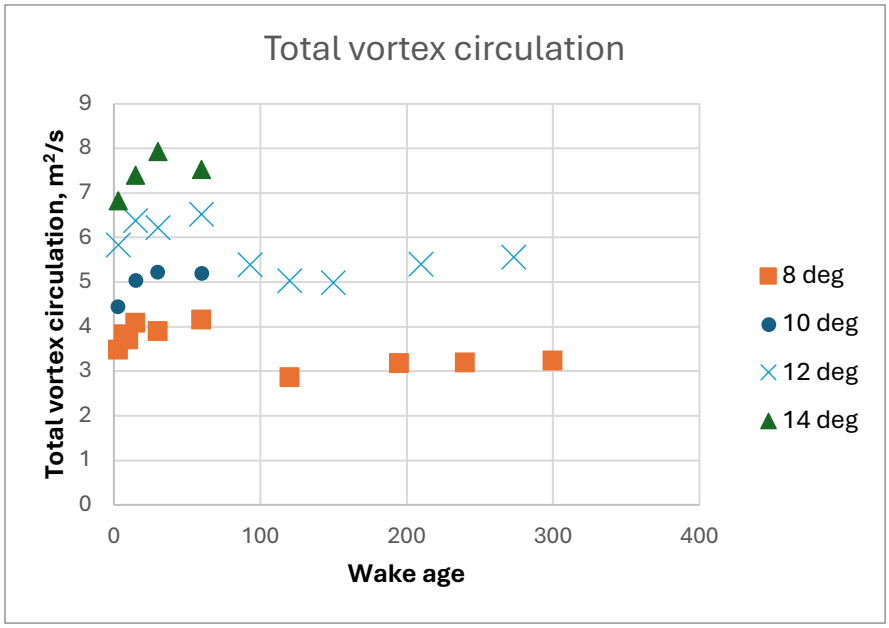
std y_v/R – standard deviation of the axial location of the tip vortex

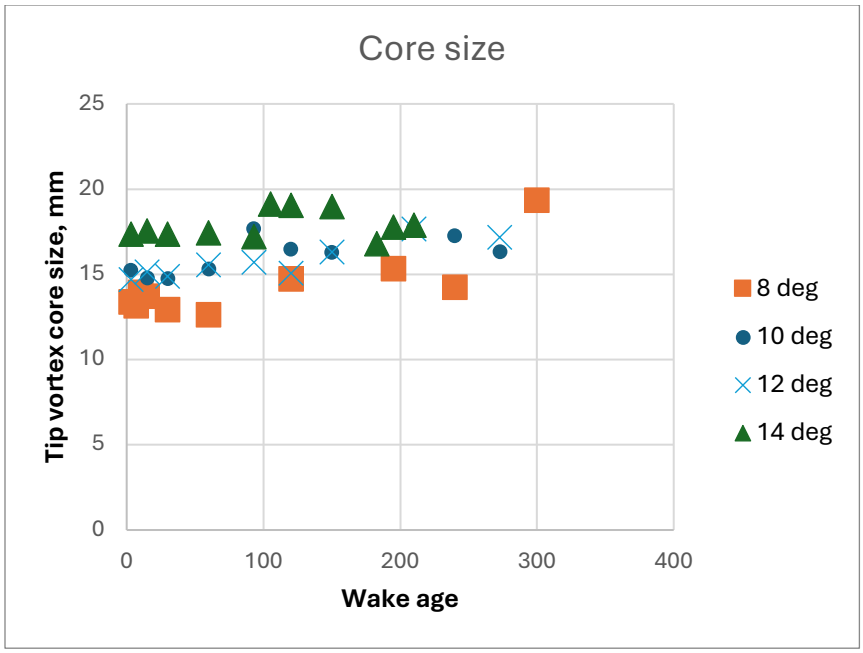
B1, B2, B3, B4 – Blades 1 to 4, respectively

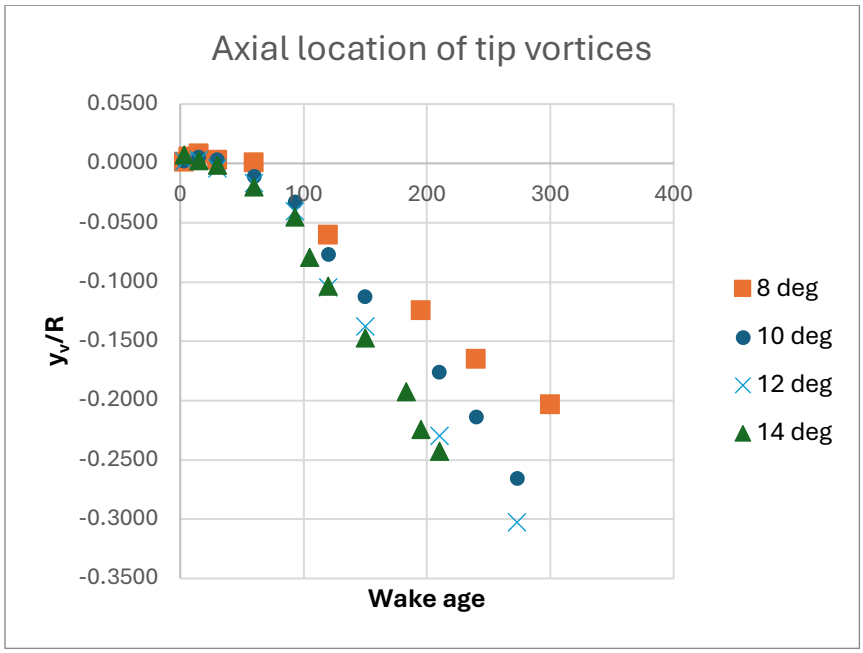
For the 8 and 12 deg collective pitch cases, measurements were made over the entire blade rotation (0 to 360 deg). For the 10 and 14 deg cases, the measurements were made for only 1/4th of the revolution (0-90 deg). So, data from each 90 degrees correspond to a different blade. Data points in the files are thus marked with the associated blade number in the last column.

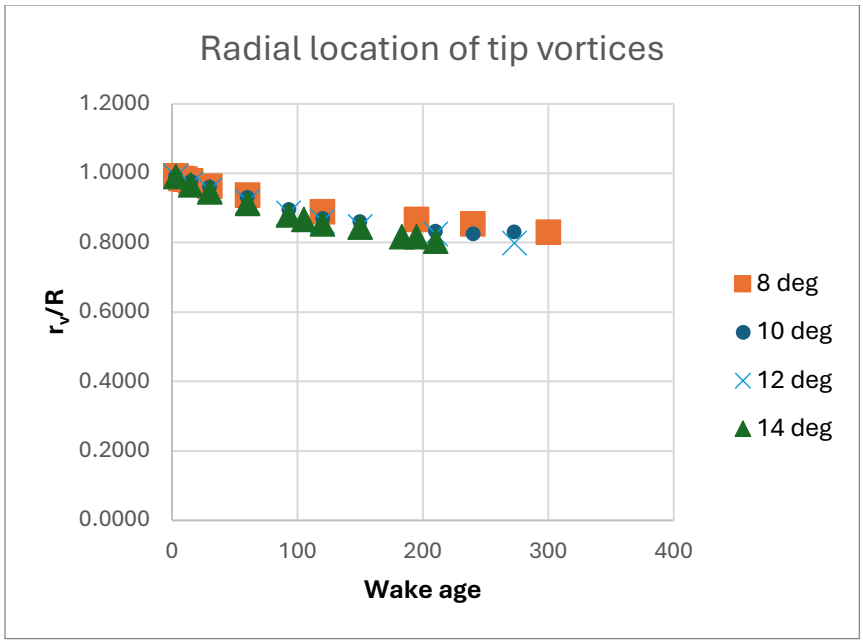
Similar to the vector field data, not all 200 realizations contributed to the calculated statistics of mean and standard deviation in the tip vortex locations, core size, or peak swirl velocity. A local criterion (matching two different methods of vortex circulation) was developed to ensure that the vector void (caused by seed void) and the presence of the vortex sheet do not contribute to the calculation of tip vortex properties (see Ref. 1).

Sample plots of the measured/extracted vortex core properties and trajectories are provided in the figures below.




Core size
Tip vortex core size, mm
25
20
15
10
5
0
0
100
200
300
400
Wake age
8 deg
10 deg
12 deg
14 deg


Axial location of tip vortices
y_v/R
0.0500
0.0000
-0.0500
-0.1000
-0.1500
-0.2000
-0.2500
-0.3000
-0.3500
0
100
200
300
400
Wake age
8 deg
10 deg
12 deg
14 deg


Radial location of tip vortices
1.2000
1.0000
0.8000
0.6000
0.4000
0.2000
0.0000
r_v/R
0
100
200
300
400
Wake age
8 deg
10 deg
12 deg
14 deg